

Introduction

Parabens are synthetic preservatives widely detected in freshwater ecosystems globally, with **methylparaben** (MeP) identified as the most prevalent

Map of Global Paraben Detection



- Parabens have been found to alter heart rate, gene expression, and behavior in aquatic organisms
- Daphnia magna*** is a common ecotoxicological model due to its high sensitivity to contaminants, short reproductive cycle, and ecological relevance as a keystone species
- DNA methylation**, where methyl groups bind to cytosine to regulate gene expression, is a likely mechanism for these effects
 - These alterations can be passed to unexposed offspring via **transgenerational inheritance**

Purpose

Current research focuses on acute MeP exposure at high concentrations. This research aims to determine the **long-term** ecological impact of MeP on aquatic organisms by measuring heart rate and DNA methylation levels in *D. magna* across generations and comparing them.

Hypothesis

As an endocrine disruptor, increasing MeP exposure impairs *D. magna* development, causing reduced parental heart rates and significant global DNA methylation changes in the F3 generation compared to controls.



A daphnid under a compound light microscope (4x magnification), taken while measuring heart rate.

Investigating the Transgenerational Effects of Methyl 4-Hydroxybenzoate Exposure on the Water Flea, *Daphnia magna*

Method

IV

- Concentration of MeP (0.5, 1, 4 mg/L)

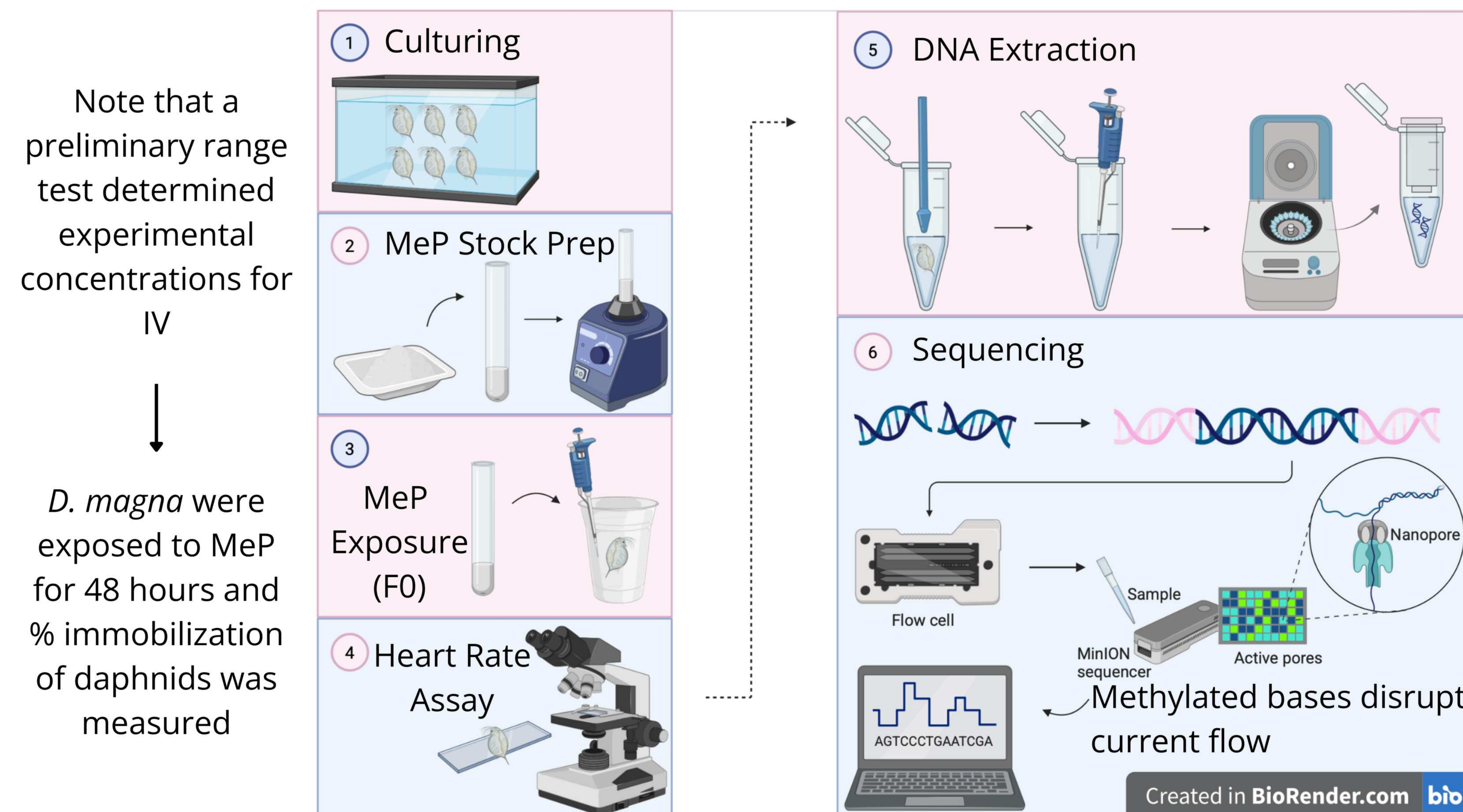
DV

- Heart rate (bpm)
- global DNA methylation levels (% 5mC)

Controls

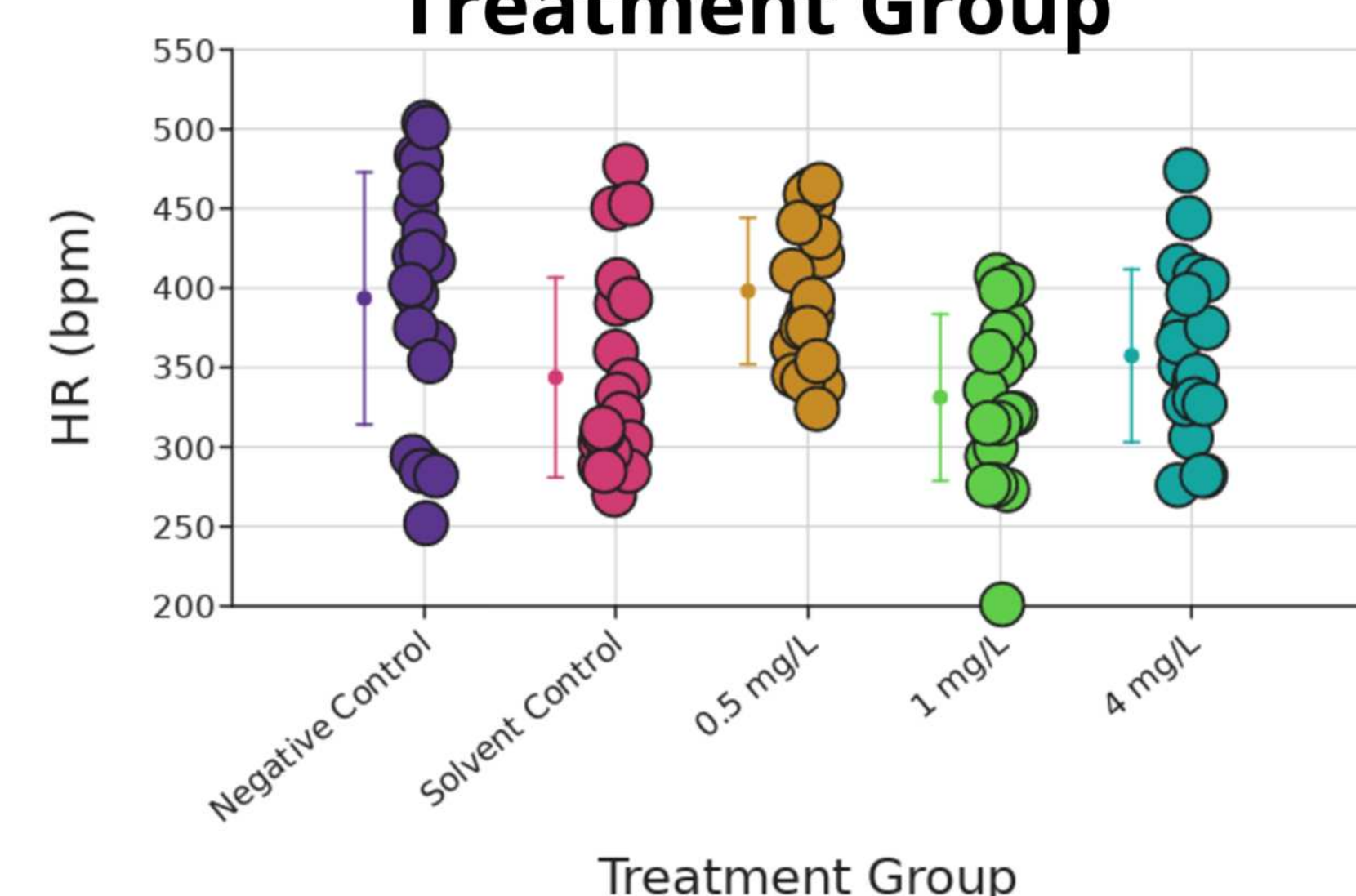
- Negative control group (no MeP, no DMSO)
- solvent control group (no MeP)

Transgenerational Experiment



Results

Heart Rate (bpm) of *D. magna* After 48 hours of MeP Exposure per Treatment Group



Kruskal-Wallis Test

$$H(4) = 16.96, p < 0.01$$

Dunn's Post-hoc Test (Bonferroni correction)

Significant differences were found between the following pairs:

- Negative Control vs. 1 mg/L ($p = 0.05$)
- Solvent Control vs. 0.5 mg/L ($p = 0.04$)
- 0.5 mg/L vs. 1 mg/L ($p = 0.01$)

No significant differences were detected between all other pairs ($p = 0.39-1.00$).

Conclusions

References

- Bolujoko, N. B., et al. (2022). Occurrence and human exposure assessment of parabens in water sources in Osun State, Nigeria. *Science of the Total Environment*, 814, 152846. <https://doi.org/10.1016/j.scitotenv.2021.152846>
- Eghan, K., Lee, S., & Kim, W.-K. (2023). Cardio- and neuro-toxic effects of four parabens on *Daphnia magna*. *Ecotoxicology and Environmental Safety*, 268(1), 115670. <https://doi.org/10.1016/j.ecoenv.2023.115670>
- Ebert, D. (2022). *Daphnia* as a versatile model system in ecology and evolution. *EvoDevo*, 13, 16. <https://doi.org/10.1186/s13227-022-00199-0>